



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	Cognix: Personal Knowledge Operating System with AI Memory	
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Abstract:

Cognix is a personal knowledge operating system designed to give AI assistants a persistent, context-aware memory of a specific user. Conventional LLM-based assistants have no memory across sessions, and standard Retrieval-Augmented Generation (RAG) pipelines are built to search static external documents rather than a user's own evolving notes, preferences, and past decisions. Passing entire conversation histories or large documents into the LLM context window is costly, runs into context-length limits, and buries the details that matter under irrelevant text. This creates a clear need for a dedicated memory layer that can store, organize, and retrieve personal knowledge efficiently.

To address this need, Cognix combines semantic search, structured knowledge representation, and adaptive context retrieval into a single memory layer that sits between the user and the underlying LLM. Three components form the core of the system.

The first component is the memory ingestion and indexing pipeline, which processes conversations, notes, and documents into embeddings and extracts key entities, storing them as structured, searchable memory units rather than flat text blocks.

The second component is a concept relationship graph that links related memory units by topic, entity, and event, letting the system traverse connections between ideas instead of retrieving isolated, unrelated fragments — addressing a key limitation of plain chunk-based RAG.

The third component is an adaptive context-compaction stage that selects and compresses only the most relevant memories for a given query into a compact prompt, keeping the context sent to the LLM both relevant and token-efficient rather than relying on a fixed top-K retrieval.

Early evaluation criteria for Cognix focus on personalization accuracy, response relevance, and token efficiency, measured against plain conversation-history replay and standard RAG baselines. The goal is to show that structured, relationship-aware memory retrieval improves both the quality and the efficiency of personalized AI interactions.

Collectively, these components form a reusable memory substrate rather than a feature tied to one assistant, positioning Cognix as infrastructure that any LLM-based application can plug into to gain a persistent, structured understanding of its users over time.

Project Architecture/Project Overview:

